

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

15 September 2014

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	13
2	
3	15
4	10
5	
6	
7	
Section B	
	20
Total	100

This document consists of ** printed pages and 1 blank page.

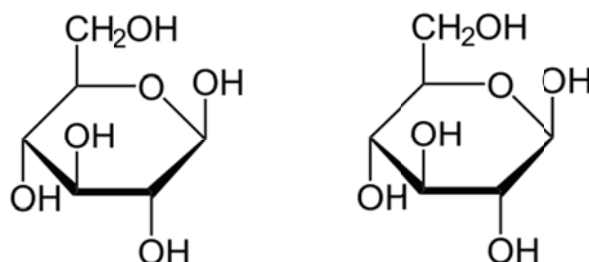


Innova Junior College

[Turn over

Section AAnswer **all** questions.

- 1 Fig. 1.1 shows two monosaccharide units that are linked by cellulose synthase to form cellulose in plant cells.

**Fig. 1.1**

- (a) (i) With reference to Fig. 1.1, name and illustrate the formation of bond between the two monosaccharide units.

***show 180° inversion of adjacent β -glucose units ;
show loss of one H₂O molecule ;
show formation of β -1,4 glycosidic bond ;
correct annotation ;***

[2]

name of bond **β -1,4 glycosidic bond** [1]

- (ii) Describe **two** structural features that enable cellulose to perform its structural function in plants.

- 1. alternate arrangement of β -glucose → allows formation of straight chains → allow bundling to form microfibrils and macrofibrils → increases tensile strength ;;***
- 2. protruding OH groups → allow formation of H-bonds → increases tensile strength ;;***
- 3. association of cellulose with hemicellulose and pectin in cell wall → increases tensile strength ;;***

Any 2 points

[2]

(b) In herbivores, such as cows, microbes present in the gut produces the enzyme cellulase. Cellulase catalyses the breakdown of cellulose into monosaccharide units which are taken into the cells in the gut by the process of facilitated diffusion. This process involves transmembrane transport proteins called GLUT.

(i) State the type of reaction catalysed by cellulase, and explain how cellulase lowers the activation energy of the breakdown of cellulose.

1. hydrolysis reaction ;;

2. Binding of substrate (cellulose) to cellulase's active site

to formation of ES complex;;

3. Induce bond stress in substrate (cellulose) (by reducing bond $\angle$)

Increase likelihood of bond breakage;;

4. Increase substrate reactivity / stabilising TS

**By altering charge and electron distribution within substrate /
providing alternative pathway;;**

5. Bring rxtants i.e. water and cellulose

in to close proximity & correct orientation;;

[4]

(ii) Describe and explain how the monosaccharide units are taken into the gut cells by facilitated diffusion.

1. β -glucose $\rightarrow$ large, polar molecule

**repelled by hydrophobic core (consisting of hydrophobic tails of
phospholipids) of cell surface membrane;;**

**2. require use of transmembrane transport proteins providing
hydrophilic channel ;;**

**3. β -glucose is of higher conc. outside cell $\rightarrow$ moves down conc
gradient (higher to lower) into cell**

4. w/o use of ATP;;

[3]

(iii) Suggest how the gut cell can reduce the number of GLUTs in its cell surface membrane.

1. Ubiquitin-targeted proteolysis leading to GLUT degradation;;

**2. Via endocytosis, packaged to form endocytic vesicle and removed
from CSM ;;**

Any one

[1]

[Total: 13]

- 2 Fig. 2.1 shows ribosomes attached to the endoplasmic reticulum, carrying out insulin synthesis.

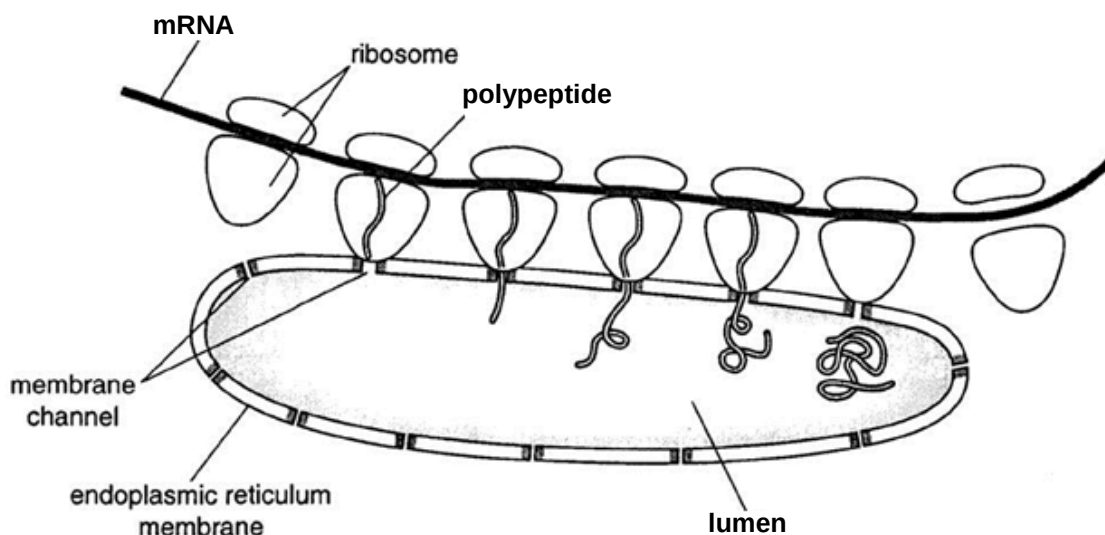


Fig. 2.1

- (a) The polypeptide entering the lumen is processed before it leaves the endoplasmic reticulum. Describe the post-translational processing of the polypeptide to form insulin.

1. signal peptide cleaved, pre-proinsulin $\rightarrow$ proinsulin;;

2. C peptide cleaved, diS bridges formed btw chain A and chain B

proinsulin folded $\rightarrow$ insulin;;

[2]

- (b) Insulin synthesised is secreted into the bloodstream. Presence of insulin in the bloodstream lowers blood glucose level. Thus production of insulin is highly regulated.

- (i) Briefly describe the events leading up to the secretion of insulin after its processing in the lumen.

1. trspt vesicle with insulin fuse with cis face of GA

& pinch off in secretory vesicle at trans face;;

2. insulin undergoes sorting, packaging & modification by adding, deleting & substituting sugar monomer oligosacc chain

to form molecular identity tag;;

3. translocate to/w CSM with aid of cytoskeleton

where vesicle fuse with CSM to release insulin into bloodstream via exocytosis;;

4. release is triggered by $\uparrow$ ATP:ADP ratio in β -cell in IoL

due to $\uparrow$ glycolysis from $\uparrow$ glc uptake;;

5. cause closure of ATP sensitive K^+ channel

leading to K^+ accumulation thus depolarisation;;

6. which triggers opening of voltage-gated Ca^{2+} channels

Ca^{2+} influx triggers release of insulin via exocytosis;;

[3]

(ii) Suggest how expression of the insulin gene can be regulated at translation level.

1. ting stability / lifespan / half-life of mRNA by ting length of polyA tail

2. protecting mRNA from enzymatic degradation in cytoplasm;;

3. gene silencing by miRNA complexed w RISC

4. which binds partially complementarily to mRNA coding for insulin to translation OR

completely complementarily to mRNA coding for insulin to degrade mRNA;;

[2]

Fig. 2.2 shows the sequence of bases in a section of the mRNA in Fig. 2.1, coding for the **first few amino acids** of a polypeptide chain.

3' AUGUUGCCCCCUUAGGUAAU 5'

Fig. 2.2

Each codon codes for a specific amino acid as illustrated in Table 2,

Table 2

UUU	phe	UCU	ser	UAU	tyr	UGU	cys
UUC	phe	UCC	ser	UAC	tyr	UGC	cys
UUA	leu	UCA	ser	UAA	STOP	UGA	STOP
UUG	leu	UCG	ser	UAG	STOP	UGG	trp
CUU	leu	CCU	pro	CAU	his	CGU	arg
CUC	leu	CCC	pro	CAC	his	CGC	arg
CUA	leu	CCA	pro	CAA	gln	CGA	arg
CUG	leu	CCG	pro	CAG	gln	CGG	arg
AUU	ile	ACU	thr	AAU	asn	AGU	ser
AUC	ile	ACC	thr	AAC	asn	AGC	ser
AUA	ile	ACA	thr	AAA	lys	AGA	arg
AUG	met	ACG	thr	AAG	lys	AGG	arg
GUU	val	GCU	ala	GAU	asp	GGU	gly
GUC	val	GCC	ala	GAC	asp	GGC	gly
GUA	val	GCA	ala	GAA	glu	GGA	gly
GUG	val	GCG	ala	GAG	glu	GGG	gly

- (c) (i) Using information from Table 2, list the corresponding amino acids coded by the mRNA in Fig. 2.2.

met – asp – ser – pro – val - val

[1]

- (ii) Explain why each codon consists of three bases.

1. 20 common aas present thus need minimum of 20 codes;;

2. if 1 base ➡ only 4 codes

if 2 bases ➡ only 16 codes

thus need 3 bases ➡ 64 codes;;

3. allow for stop codes to terminate translation;;

[2]

A ten amino acid long section of the polypeptide chain was partially hydrolysed to release oligopeptides shown in Fig. 2.3.

gly-ala-gly-val

gly-gly-lys

gly-val-ile-phe

phe-thr-gly-gly

Fig. 2.3

- (d) With reference to Fig. 2.3 work out the order of amino acids in the polypeptide.

gly-ala-gly-val-ile-phe-thr-gly-gly-lys ;;

[1]

- (e) (i) Explain why single base deletions in a coding DNA sequence are more likely to produce non-functional proteins than single base substitutions.

1. single base deletions ➡ frameshift mutation

all aas downstream of mutation will be Δ ;;

2. single base sub only affects 1 codon thus 1 aa

i.e. missense mutation $\Rightarrow$ less Δ to primary struct of prot;;

3. silent mutation if sub occurs at 3rd base of codon

to result in degenerate code coding for the same aa;;

® nonsense mutation as a cf as substitution may also lead to stop code

[2]

- (ii) Cystic fibrosis (CF) results from base deletions in the CFTR gene. Explain why the consequence is different from usual single base deletions.

1. (Δ F508) involves deletion of 3 bases $\Rightarrow$ $\therefore$ no frameshift;;

2. leading to deletion of single aa, phenylalanine;;

[2]

[Total: 15]

- 3** The Middle East Respiratory Syndrome (MERS) is a fatal viral respiratory illness caused by a strain of coronavirus called MERS-CoV. Through its spike proteins, the virus infects cells in the respiratory tract in humans, leading to symptoms of fever, cough and shortness of breath, and eventually death.

Fig. 3.1 shows a model of the MERS-CoV.

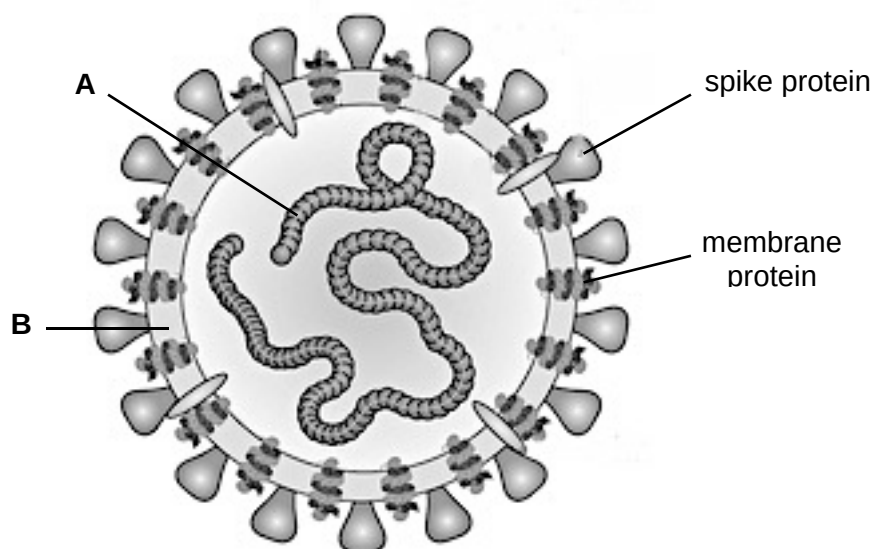


Fig. 3.1

- (a) (i) Identify structure A.

nucleocapsid;; ® capsid/genome/RNA/nucleic acid

[1]

(ii) Explain the role of the positive sense RNA present in structure A.

1. Acts as the mRNA/template strand for translation

to synthesize viral proteins/enzymes (required in production/assembly of new virions) ;;

2. Acts as template to form negative sense RNA

which is used as a template to replicate new positive sense RNA viral genome ;;

[2]

(iii) Describe how MERS-CoV acquires structure B.

1. Spike proteins and membrane proteins are synthesised at the rER

inserted into the ER membrane of host cell ;;

2. translocated via vesicles to host cell CSM

cell surface proteins are inserted via fusion of vesicle membrane with CSM ;;

3. assembled viral core travels to same site and is released via budding

hence virus acquires structure B (outer envelope) from CSM of the host cell ;;

[3]

(b) A vaccine for MERS-CoV is to be developed in the near future. However, scientists warn that vaccines produced against the strain of MERS-CoV may not prevent future epidemics from occurring.

Explain why this is so.

1. Antigenic drift

due to lack of proofreading mechanism by RNA polymerase ;;

2. random accumulation of mutations in genome

results in gradual change in structure of antigens/spike proteins ;;

3. i.e. specific antibodies produced by B lymphocytes after receiving vaccine targeting specific MER-CoV strain

will not be able to recognise and bind to new antigens → no immune response ;;

® antigenic shift

[3]

(c) Unlike MERS-CoV, some human viruses are able to integrate their genome into the host cell genome. This sometimes leads to the development of cancer.

With a named example, explain how a virus may lead to the development of cancer.

1. Human Immunodeficiency Virus (HIV)

insertional mutagenesis may occur;;

2. viral genome randomly integrates via integrase

into the host cell genome to form a provirus ;;

3. GOF of proto-oncogene → oncogene ;

resulting in excessive stimulation of cell division ;;

4. LOF mutation in TSG ;

resulting in inability to halt cell cycle / repair damaged DNA ;;

5. Activation of telomerase gene ;

resulting in maintenance of/preventing shortening of telomeres → allows indefinite DNA replication ;;

6. Activation of genes for angiogenesis ;

promotes formation of blood vessels to supply tumour cells with nutrients and oxygen ;;

7. Activation of genes involved in metastasis and tissue invasion ;

allows cancer cells to enter blood stream / lymph and invade tissues ;;

Must 1 – 2, any two other pairs

[4]

- (d) Scientists have sequenced the genome of MERS-CoV, and through bioinformatics analysis they have identified a low variable RNA segment in its genome. This allowed the synthesis of a short double-stranded RNA called the small interfering RNA (siRNA), which when introduced into the infected cell, is capable of preventing viral genome replication and protein synthesis.

Suggest how the siRNA is able to do so.

1. siRNA complexes with RNA-inducing silencing complex (RISC);;

2. siRNA binds to its target viral RNA segment (via complementary base-pairing) leading to degradation of viral genome ;;

[2]

[Total: 15]

- 4** A female carrier of X-linked SCID with blood type AB married a normal male with blood type O.

If in blood transfusion, only individuals without genetic diseases are able to donate their blood,

- (a) draw a genetic diagram to identify the offspring who are possible candidates for blood transfusion to their parents.

1. Correct legend ;;

2. Correct parental genotypes ;

3. Correct gametes ;

4. Correct punnet sq ;;

5. **Correct offspring genotypes ;**
6. **Correct offspring phenotypes ;**
7. **Correct candidates identified ;;**

[5]

- (b) The ABO blood type shows discontinuous variation while body mass shows continuous variation in the human population.

Describe **two** differences between the two types of variation.

1. **discont. → shows distinct / discrete phenotypes whilst cont. → shows a continuous range of phenotypes ;;**
2. **discont. → controlled by one or a few gene loci whilst cont. → controlled by many genes at diff. gene loci / polygenic ;;**
3. **discont. → small / no environmental influence on phenotype whilst cont. → large effect ;;**

must show comparison for mark to be given, any 2

[2]

- (c) In an investigation, the cross between blood type AB and blood type O individuals was analysed. A chi-square analysis was carried out and the chi-square value calculated was **1.10**. Using chi-squared values in **Table 4**, draw appropriate conclusions as to whether the results of the cross actually followed the expected ratio. Include in your conclusions a proposed null hypothesis.

Table 4

D.O.F	Probability, p				
	10%	5%	2%	1%	0.1%
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

1. At $p = 0.05$ and degrees of freedom = 1, critical $\chi^2 = 3.84$

Calculated χ^2 is 1.1 and is less than critical χ^2 value ;;

2. Therefore the deviation between the observed and expected ratios is statistically insignificant

the difference is due to chance alone ;;

3. Hence, H_0 is not rejected

and the observed results follows the 1:1 ratio ;;

[3
]

[Total: 10]

- 5 Fig. 5 shows a neurone, neurone Z, forming synapses with adjacent neurones, neurone X and Y.

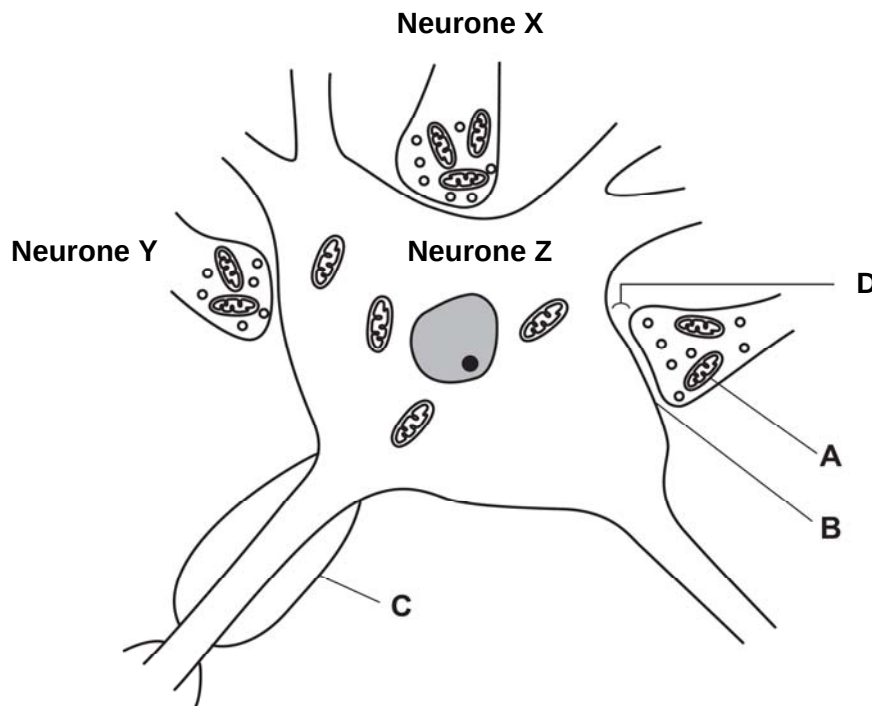


Fig. 5

- (a) (i) Name structures A to D.

A **mitochondrion; ® mitochondria**

B **post synaptic membrane;**

C *myelin sheath / Schwann cell;*

D *synaptic cleft / gap; ® synapse*

[4]

(ii) Outline the role of structure C in nerve impulse transmission.

1. *act as electrical insulator by preventing Na^+ and K^+ exchange / move^m*

allows AP to be generated only at nodes of Ranvier / impulse jump from node to node;;

2. *speeds up transmission*

by saltatory conduction;;

NB: allow combination of pts

[2]

(b) Neurone X releases acetylcholine, a neurotransmitter which stimulates the production of nerve impulses in neurone Z. Neurone Y releases GABA, another neurotransmitter which results in the inhibition on the production of nerve impulses in Neurone Z.

(i) Describe the sequence of events, from the arrival of a nerve impulse to neurone X, leading to the release of acetylcholine.

1. *arrival of impulse cause opening of voltage-gated Ca^{2+} channels*

leading to Ca^{2+} influx ➔ ↑ in intracellular $[\text{Ca}^{2+}]$;;

2. *triggers translocation of synaptic vesicles to/w the presynaptic mem*

vesicle fuses w mem ➔ ACh is released into the synaptic cleft via exocytosis;;

[2]

(ii) Explain how Neurone Y inhibits the production of nerve impulses in neurone Z.

1. *release of GABA into synaptic cleft binds & opens ligand-gated K^+ OR Cl^- channels*

leading to K^+ efflux OR Cl^- influx fr postsynaptic neurone;;

2. *leading to hyperpolarisation < -70mV*

generates IPSP;;

[2]

(c) Epilepsy is a neurological disorder resulting in seizures when there is abnormal increase in neurone activity in the brain. Epilepsy has been linked to an abnormality in GABA receptors.

Suggest and explain how an abnormality in GABA receptors may result in epilepsy.

1. *△ in 3D conformation of GABA receptors on postsynaptic mem*

⊗ complementary to GABA ➔ ⊗ bind GABA;;

2. *impulse from neurone Y unable to inhibit EPSP from neurone X (by temporal summation)*

leading to over stimulation / too many EPSP generated in neurone Z
due to ACh release fr X;;

[2]

[Total: 12]

- 6 The rosewood, *Tipuana tipu*, and the rambutan, *Nephelium lappaceum*, are some plants that share a common ancestor. Fig. 6 shows the variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.

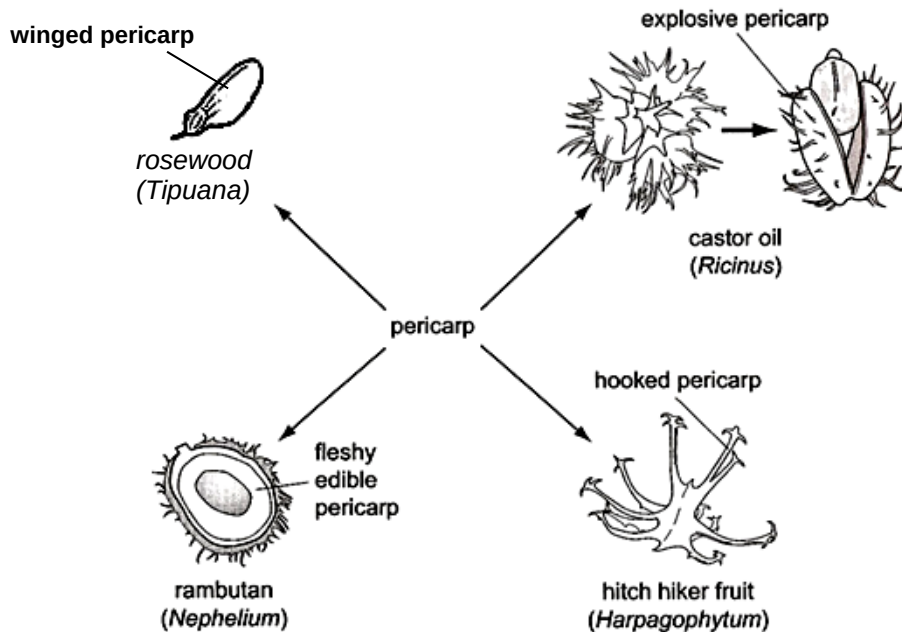


Fig. 6

- (a) Complete Table 6 to show the classification of the rosewood.

Table 6

kingdom	<i>Plantae</i>
<i>phylum</i>	Streptophyta
<i>class</i>	Magnoliopsida
order	Fabales
<i>family</i>	Fabaceae
genus	<i>Tipuana</i>
<i>species</i>	<i>Tipuana tipu</i>

[3]

- (b) With reference to Fig. 6, explain how natural selection can lead to speciation.

1. **Presence of variation in pericarp in single common ancestor pop ;;**

2. **Presence of different biotic and abiotic conditions in diff env niche leads to different selection pressures**

reference to fig. 6.1 to describe eg of selection pressure ;;

- 3. Plants with adaptations suited to selection pressure will be selected for/selective adv;;**
- 4. i.e. higher survival rate & reproductive rate (i.e. reproductive success)
i.e. alleles/genes of selected adaptations will be passed on to their offspring ;;**
- 5. over time, reproductive barriers (pre-, post-zygotic) may form between populations
preventing interbreeding and thus no gene flow occurs ;;**

[5]

- (c) Before the use of molecular homology, anatomical homology was commonly used to determine the evolutionary relationship between different species of organisms. One example of anatomical homology is homologous structures such as the pericarp seen in Fig. 6.

- (i) Explain what are 'homologous structures'.

- 1. similar structures in different spp**

that has diff functions ;;

- 2. derived from common ancestor ;**

due to divergent evolution ;;

[2]

- (ii) Explain why molecular homology is better than anatomical homology in determining evolutionary relationship.

- 1. Molecular homology compares differences/similarities between DNA/ amino acid seq ;**

- 2. Anatomical homology compares differences/similarities between morphological structures ;**

- 3. Molecular homology → objective, unambiguous**

as all organisms have 4 universal bases, A, T, C, G ;;

- 4. vs anatomical homology difficult to distinguish objectively (i.e. subjective)**

and could have arose through convergent evolution i.e. analagous structures ;;

- 5. Molecular homology → quantifiable**

allows inference of degree of relatedness between organisms that are morphologically very different/similar ;;

- 6. Molecular mtds offer a limitless set of characters to be studied**

where each nucleotide/amino acid position can be considered a character to distinguish between spp ;;

- 7. Also, rate of mutation of neutral alleles can be calculated to calibrate a molecular clock**

where the number of neutral mutations between spp can indicate the time of divergence;;

- 8. AVP**

[4]

- (iii) In the case of extinct species, molecular homology may be less effective in determining evolutionary relationships.

Suggest why this is so.

1. degradation of nucleic acids ;

leading to incomplete sequences i.e. difficulty in comparison ;;

2. AVP;;

[1]

[Total: 15]

Section B

- 7 (a) Describe the organisation of DNA in a chromosome during mitosis. [6]
1. ***DNA double helix is coiled $1\frac{3}{4}$ turns/ 146 bp around octamer of (8) histones forming the nucleosome core;;***
 2. ***ionic interxns btw -vely charged DNA & +vely charged histone stabilises the nucleosome core;;***
 3. ***further stabilised by a H1 histone forming a complete nucleosome joined by spacer DNA to form a 10nm 'beads-on-string' structure;;***
 4. ***further coiled into solenoid forming 30 nm chromatin fibre;;***
 5. ***during prophase, DNA loops with the aid of scaffold proteins and spiralise / condense into more compact form;***
 6. ***appears as metaphase chromosome consisting of a pair of sis chromatids held at centromere;;***
- (b) Outline the events of mitosis. [8]
- Prophase***
1. ***chromatin undergoes condensation / spiralisation to become visible as chromosomes;;***
 2. ***nuclear membrane fragments, nucleolus disappear;;***
 3. ***centrioles move to oppo ends of the cell, short microtubules organise around centrosome to form aster;;***
 4. ***microtubules elongate (via polymerisation of tubulin dimers) to form spindle fibre;;***
- Metaphase***
5. ***kinetochore spindle fibres attached to the kinetochore at the centromere of chromosomes to (randomly) align them at metaphase plate;;***
- Anaphase***
6. ***centromere divides & kinetochore spindle fibres shorten to pull sister chromatids to oppo pole of the cell led by its centromere;;***
 7. ***non-kinetochore spindle fibres interact to lengthen the cell;;***
- Telophase***
8. ***chromosomes uncoil / decondensed to become chromatin threads;;***
 9. ***spindle fibres disintegrate, nuclear envelope reforms around the chromosomes at each pole & nucleoli reorganise;;***
 10. ***(cytoplasm divides) in cytokinesis (by cleavage furrow / cell plate) to give 2 genetically identical daughter cells;;***
- (c) Describe the similarities and differences of homologous chromosomes. [6]
1. ***are nearly identical in size;;***
 2. ***same gene loci i.e. same gene at the same position***
 3. ***same banding pattern;;***
 4. ***same centromere location;;***
 5. ***may have different alleles for the same gene;;***
 6. ***different origin of inheritance i.e. one from paternal, the other maternal;;***

[Total: 20]

8 (a) Compare the endocrine and nervous system.

[5]

	Endocrine System	Nervous System
1. messenger	by chemical messengers	by electrical impulse along axon chemical neurotransmitters across synapse
2. path of transmission	via blood circulation	via neurones
3. speed of transmission	relatively slow	rapid
4. speed of response	slow	rapid
5. duration of effect	often longer lasting	response short-lived
6. coverage of effects	widespread to all target organs with appropriate receptors,	localised effects, only to group of innervated effectors
7. permanence	may be permanent & irreversible	often temporary & usually reversible

(b) Describe how an action potential is transmitted along a myelinated neurone.

[8]

- 1. arrival of stimulus at resting neurone $\Rightarrow$ some voltage-gated Na^+ channels open $\Rightarrow$ Na^+ diffuses into axon (down its electrochemical gradient) causing initial depolarisation;;**
- 2. initial depol $\geq$ threshold value of -50 mV $\Rightarrow$ positive feedback occurs leading to opening of more voltage-gated Na^+ channels;;**
- 3. rapid influx of Na^+ leads to a depolarisation to $+40 \text{ mV}$, generating an AP;;**
- 4. voltage-gated Na^+ channels close & are inactivated, axon is in absolute refractory period;;**
- 5. voltage-gated K^+ channels open leading to the efflux of K^+ (down its electrochemical gradient) $\Rightarrow$ repolarised to/w RP of -70 mV ;;**
- 6. voltage-gated Na^+ channels regain activity, axon is in relative refractory period;;**
- 7. voltage-gated K^+ channels begin to close, but slow closing of channels allow an excess of K^+ to leave the axon $\Rightarrow$ hyperpolarisation of -90 mV ;;**
- 8. RP is re-established by differential rates of (facilitated) diffusion of K^+ and Na^+ ions through the 'leak' channels;;**
- 9. Na^+ diffuse to adjacent node of Ranvier causes depolarisation to generate AP at next node (setting up a local current);;**
- 10. saltatory conduction, impulse transmission is unidirectional;;**

(c) With reference to signal transduction, outline how insulin brings about a lowering of blood glucose concentration.

[7]

- 1. binding of (2) insulin molecules to ligand binding site on RTK on liver cell results in a conformational Δ to RTK dimer;;**
- 2. tyrosine kinase of each receptor monomer phosphorylates tyr residues in the cytosolic domain of its dimer partner**

via autophosphorylation / cross phosphorylation;;

3. *phosphotyrosine residues serve as docking ports for adaptor protein / IRS1;;*
4. *IRS-1 phosphorylate relay proteins / SH2 domain-PI3 kinase / protein kinases which go on to phosphorylate other proteins;;*
5. *triggering phosphorylation cascade leading to signal transduction / being transduced to a series of cellular responses;;*
6. *e.g. GLUT translocation to CSM to ↑ cell's permeability to glucose thus ↑ glucose uptake by cell from blood stream;;*
7. *↑ activity of glycolytic enzymes thus use of glucose as respiratory substrate;;*
8. *activation of glycogen synthase to ↑ usage of glucose in glycogenesis;;*
9. *inhibition of glycogen phosphorylase to ↓ production of glucose via glycogenolysis;;*
10. *AVP;;*

[Total: 20]